

# SIMATS ENGINEERING

## CO3 - ASSESSMENT TOOL 2

### SIMULATION TASK

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## DIGITAL SIGNAL PROCESSING

### 20. Decimation by Factor 4

Simulate decimation by a factor of 4 with and without anti-alias filtering and analyze aliasing effects. (25) (BL4) (WK4)

Given Data:

- Sampling Frequency = 16 kHz
- Decimation Factor = 4

Tasks:

1. Implement decimation with anti-alias filtering.
  2. Repeat without filtering.
  3. Compare output spectra.
  4. Identify aliasing frequencies.
- 
5. Measure distortion.
  6. Explain the results.

## MATLAB Implementation

### Decimation WITH Anti-Alias Filtering

An FIR low-pass filter (Hamming window, order 100, cutoff =  $F_{s\_new}/2 = 2$  kHz) is designed and applied to the signal before down-sampling by  $M = 4$ :

```
%% Decimation by Factor 4 WITH Anti-Alias Filtering
% CO3 - AT2 Simulation Task 20
clc; clear; close all;

%% Given Data
Fs = 16000; % Original sampling frequency (Hz)
M = 4; % Decimation factor
FsNew = Fs / M; % New sampling frequency = 4000 Hz
NyqNew = FsNew / 2; % New Nyquist frequency = 2000 Hz

%% Test signal: in-band tones (survive) + out-of-band tones (would alias)
t = 0:1/Fs:0.05-1/Fs;
f_inband = [800 1800]; % Hz - inside new Nyquist band
f_outband = [3000 4500 6500]; % Hz - outside new Nyquist band

x = sin(2*pi*f_inband(1)*t) + sin(2*pi*f_inband(2)*t) + ...
    0.8*sin(2*pi*f_outband(1)*t) + 0.8*sin(2*pi*f_outband(2)*t) + ...
    0.8*sin(2*pi*f_outband(3)*t);

%% Step 1: Design Anti-Alias Low-Pass FIR Filter (cutoff = NyqNew)
Norder = 100;
Wn = NyqNew / (Fs/2); % normalized cutoff
```

```

h = fir1(Norder, Wn, hamming(Norder+1));

%% Step 2: Apply anti-alias filter BEFORE downsampling
x_filtered = filter(h, 1, x);

%% Step 3: Downsample by M (decimation)
y_with_filter = downsample(x_filtered, M);

% Equivalent single-line MATLAB command using Signal Processing Toolbox:
% y_with_filter = decimate(x, M, 'fir');

%% Step 4: Plot spectra
NFFT = length(x);
f_axis = (0:NFFT/2-1)*(Fs/NFFT);
Xf = abs(fft(x, NFFT)); Xf = Xf(1:NFFT/2);

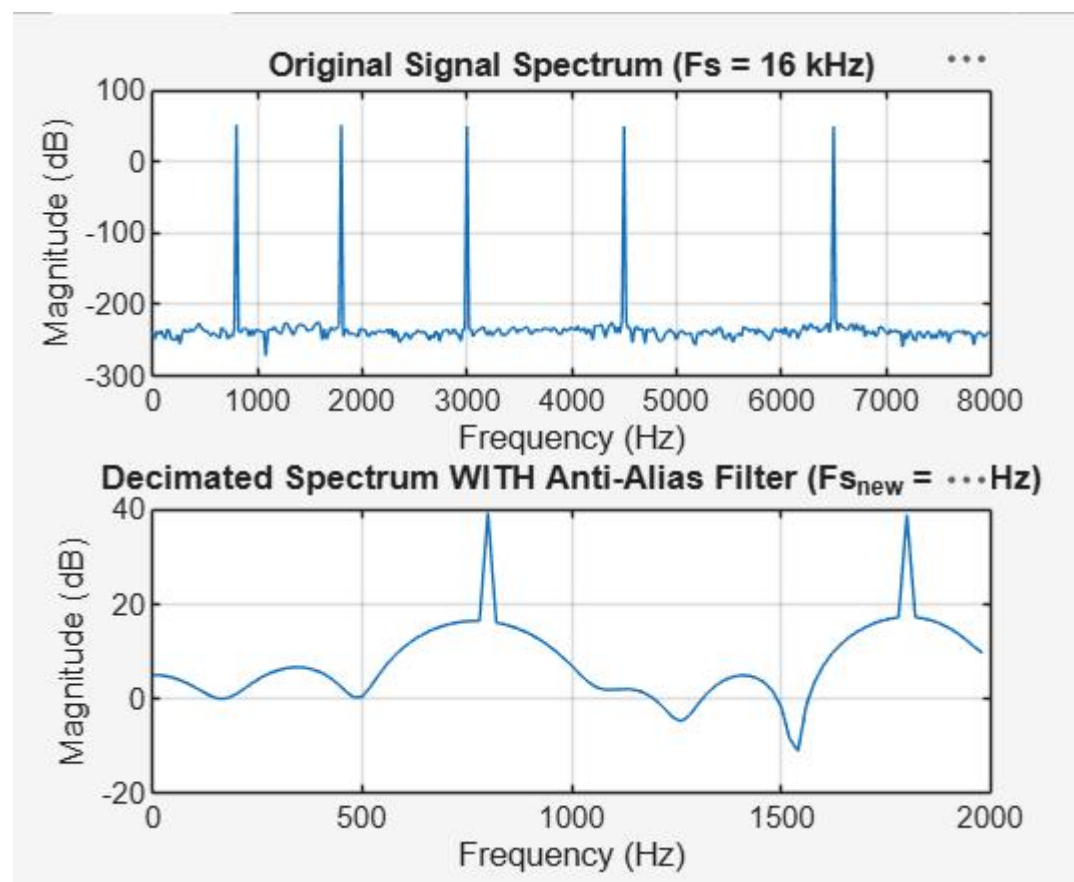
NFFT2 = length(y_with_filter);
f_axis2 = (0:NFFT2/2-1)*(FsNew/NFFT2);
Yf = abs(fft(y_with_filter, NFFT2)); Yf = Yf(1:NFFT2/2);

figure;
subplot(2,1,1); plot(f_axis, 20*log10(Xf+eps)); grid on;
title('Original Signal Spectrum (Fs = 16 kHz)');
xlabel('Frequency (Hz)'); ylabel('Magnitude (dB)');

subplot(2,1,2); plot(f_axis2, 20*log10(Yf+eps)); grid on;
title('Decimated Spectrum WITH Anti-Alias Filter (Fs_{new} = 4 kHz)');
xlabel('Frequency (Hz)'); ylabel('Magnitude (dB)');

```

## OUTPUT:



## 5.2 Decimation WITHOUT Anti-Alias Filtering (Naive)

The same signal is directly down-sampled by  $M = 4$  without any pre-filtering, and the theoretical alias frequencies are computed for comparison:

```
%% Decimation by Factor 4 WITHOUT Anti-Alias Filtering (Naive Downsampling)
% CO3 - AT2 Simulation Task 20
clc; clear; close all;

%% Given Data
Fs = 16000;
M = 4;
FsNew = Fs / M;
NyqNew = FsNew / 2;

%% Same test signal as the filtered case
t = 0:1/Fs:0.05-1/Fs;
f_inband = [800 1800];
f_outband = [3000 4500 6500];

x = sin(2*pi*f_inband(1)*t) + sin(2*pi*f_inband(2)*t) + ...
    0.8*sin(2*pi*f_outband(1)*t) + 0.8*sin(2*pi*f_outband(2)*t) + ...
    0.8*sin(2*pi*f_outband(3)*t);

%% Directly downsample WITHOUT low-pass filtering
y_without_filter = downsample(x, M);

%% Predict alias frequencies for the out-of-band tones
% f_alias = | f - round(f/FsNew)*FsNew |, folded into [0, NyqNew]
alias = zeros(size(f_outband));
for k = 1:length(f_outband)
    f = f_outband(k);
    m = round(f/FsNew);
    fa = abs(f - m*FsNew);
    if fa > FsNew/2
        fa = FsNew - fa;
    end
    alias(k) = fa;
end
disp('Out-of-band tone (Hz) -> Predicted alias frequency (Hz):');
disp([f_outband(:) alias(:)]);

%% Plot spectrum
NFFT2 = length(y_without_filter);
f_axis2 = (0:NFFT2/2-1)*(FsNew/NFFT2);
Yf = abs(fft(y_without_filter, NFFT2)); Yf = Yf(1:NFFT2/2);

figure;
plot(f_axis2, 20*log10(Yf+eps)); grid on;
title('Decimated Spectrum WITHOUT Anti-Alias Filter - Aliasing Present');
xlabel('Frequency (Hz)'); ylabel('Magnitude (dB)');
```

OUTPUT:

### Command Window

```
Out-of-band tone (Hz) -> Predicted alias frequency (Hz):
    3000        1000
    4500         500
    6500        1500
```

